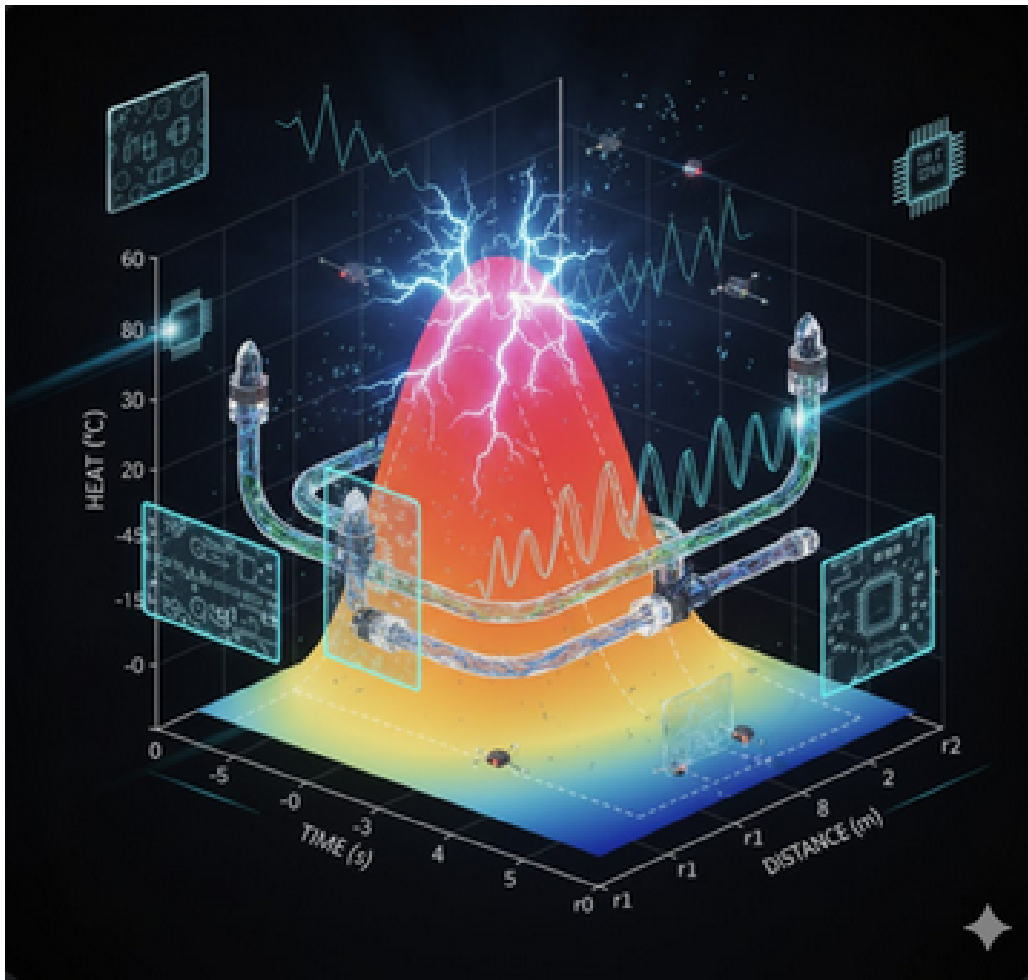


Transient Thermal Model for Insulated Cables: Theory and Examples



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List of Parameters and Units

Symbol	Description	Unit
$T_A(t)$	Conductor Temperature (Node A)	$^{\circ}$
$T_B(t)$	Insulation Surface Temperature (Node B)	$^{\circ}$
T_C	Ambient Temperature (Node C)	$^{\circ}$
T_i	Initial Temperature (at $t = 0$)	$^{\circ}$
$T_{ss,heat}$	Steady-State Temp. (Heating)	$^{\circ}$
$T(r,t)$	Temperature in Insulation (at r, t)	$^{\circ}$
$T_{soil}(r,t)$	Temperature in Soil (at r, t)	$^{\circ}$
T_{max}	Max. Insulation Temperature	$^{\circ}$
T_f	Film Temperature (for air props)	$^{\circ}$
r	General Radius	
r_1	Conductor Radius	
r_2	Insulation Outer Radius	
D	Cable Outer Diameter ($2r_2$)	
L	Cable Length	
z	Burial Depth	
A_c	Conductor Cross-Sectional Area	2
A_s	Cable Outer Surface Area	2
V_c	Conductor Volume	3
V_{ins}	Insulation Volume	3
P	Load Power	
V	Load Voltage	
PF	Power Factor	dimensionless
I	Current	
I_{max}	Ampacity (Max. Current)	
R_{elec}	Electrical Resistance	Ω
ρ_e	Electrical Resistivity	Ω
σ	Electrical Conductivity	Ω^{-1}
P_{gen}	Generated Heat (Joule Loss)	
P_{loss}	Lost Heat (to environment)	
t	Time	
τ	System Time Constant	
$C_{th,c}$	Conductor Thermal Capacitance	-1
$C_{th,ins}$	Insulation Thermal Capacitance	-1
$C_{th,corrected}$	Corrected Total Capacitance	-1
R_{ins}	Insulation Thermal Resistance	-1
R_{env}	Environmental Thermal Resistance	-1
R_{soil}	Soil Thermal Resistance	-1
R_{conv}	Convection Thermal Resistance	-1
R_{total}	Total Thermal Resistance	-1
m_c	Mass of Conductor	
m_{ins}	Mass of Insulation	
ρ_c, ρ_{ins}	Density	-3
$c_{p,c}, c_{p,ins}$	Specific Heat Capacity	-1 -1
$k_{ins}, k_{soil}, k_{air}$	Thermal Conductivity	-1 -1

Continued on next page

Table 1 – continued from previous page

Symbol	Description	Unit
ρ_{soil}	Soil Thermal Resistivity	-1
h_c	Convective Heat Transfer Coeff.	$-2 -1$
v_{wind}	Wind Speed	-1
α	Thermal Diffusivity	$2 -1$
ν	Kinematic Viscosity	$2 -1$
β	Thermal Expansion Coefficient	-1
g	Gravitational Acceleration	-2
Nu	Nusselt Number	dimensionless
Pr	Prandtl Number	dimensionless
Re	Reynolds Number	dimensionless
Gr	Grashof Number	dimensionless
Ra	Rayleigh Number	dimensionless

1 The Core Model: A Corrected ODE

The model is based on the Lumped Capacitance Method, where the cable's thermal mass is "lumped" into the conductor (T_A). This simplification reduces a complex Partial Differential Equation (PDE) to a single First-Order Ordinary Differential Equation (ODE), which is robust and solvable.

The model is governed by one fundamental energy balance:

$$C_{th,corrected} \frac{dT_A}{dt} = P_{gen} - P_{loss} \quad []$$

Where:

- $C_{th,corrected}$ (Corrected Thermal Capacitance): This is a key improvement to increase accuracy, accounting for the conductor and a fraction of the insulation's thermal mass.

$$C_{th,corrected} = (m_c c_{p,c}) + 0.5 \cdot (m_{ins} c_{p,ins}) \quad [^{-1}] \quad (1)$$

- P_{gen} (Heat Generation): The heat from Joule's Law in the conductor.

$$P_{gen} = I^2 R_{elec} \quad [] \quad (2)$$

- P_{loss} (Heat Loss): The heat escaping from the conductor to the environment.

$$P_{loss} = \frac{T_A(t) - T_C}{R_{total}} \quad [] \quad (3)$$

- R_{total} (Total Thermal Resistance): The sum of all series resistances.

$$R_{total} = R_{ins} + R_{env} \quad [^{-1}] \quad (4)$$

2 The Two Scenarios: Heating & Cooling

Both scenarios are solved using the Time Constant (τ), which defines the characteristic speed of the thermal response.

$$\tau = R_{total} \cdot C_{th,corrected} \quad [] \quad (5)$$

2.1 Scenario 1: Heating (Current I is ON)

The conductor temperature $T_A(t)$ rises from an initial temperature T_i (e.g., ambient T_C) to a final steady-state temperature $T_{ss,heat}$.

- Steady-State Temp ($T_{ss,heat}$): This is the final temperature where $P_{gen} = P_{loss}$.

$$T_{ss,heat} = T_C + (P_{gen} \cdot R_{total}) \quad [^{\circ}] \quad (6)$$

- Transient Equation:

$$T_A(t) = T_{ss,heat} + (T_i - T_{ss,heat}) \left(-\frac{t}{\tau} \right) \quad [^{\circ}] \quad (7)$$

2.2 Scenario 2: Cooling (Current is OFF)

The conductor temperature $T_A(t)$ falls from an initial hot temperature T_i (e.g., $T_{ss,heat}$) to the ambient temperature T_C . Here, $P_{gen} = 0$.

- Steady-State Temp ($T_{ss,cool}$): T_C

- Transient Equation:

$$T_A(t) = T_C + (T_i - T_C) \left(-\frac{t}{\tau} \right) \quad [^{\circ}] \quad (8)$$

3 Environmental Cases (Calculating R_{total})

The total resistance R_{total} depends on the fixed insulation resistance R_{ins} and the variable environmental resistance R_{env} .

- Insulation Resistance (Fixed): Radial conduction through a cylinder.

$$R_{\text{ins}} = \frac{(r_2/r_1)}{2\pi k_{\text{ins}} L} \quad [^{-1}] \quad (9)$$

3.1 Case 1: Underground (Conduction)

Heat conducts through the soil. R_{total} is a constant.

$$R_{\text{env}} = R_{\text{soil}} = \frac{(4z/D)}{2\pi k_{\text{soil}} L} \quad [^{-1}] \quad (\text{for burial depth } z) \quad (10)$$

This model can be solved directly.

3.2 Case 2: Overhead (Natural Convection)

Heat convects into still air. R_{total} is temperature-dependent and must be found iteratively at steady-state.

$$R_{\text{env}} = R_{\text{conv}} = \frac{1}{h_c A_s} \quad \text{where} \quad h_c = \frac{k_{\text{air}}}{D} Nu_{\text{nat}} \quad [^{-1}] \quad (11)$$

The Rayleigh Number (Ra) is calculated as $Ra = Gr \cdot Pr$.

- Churchill-Chu Equation (for Nu_{nat}):

$$Nu_{\text{nat}} = \left\{ 0.60 + \frac{0.387 Ra^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}} \right\}^2 \quad [\text{dimensionless}] \quad (12)$$

3.3 Case 3: Overhead (Forced Convection)

Heat convects into wind. R_{total} is also temperature-dependent and found iteratively.

$$R_{\text{env}} = R_{\text{conv}} = \frac{1}{h_c A_s} \quad \text{where} \quad h_c = \frac{k_{\text{air}}}{D} Nu_{\text{forced}} \quad [^{-1}] \quad (13)$$

The Reynolds Number (Re) is calculated as $Re = \frac{v_{\text{wind}} D}{\nu}$.

- Churchill-Bernstein Equation (for Nu_{forced}):

$$Nu_{\text{forced}} = 0.3 + \frac{0.62 Re^{1/2} Pr^{1/3}}{[1 + (0.4/Pr)^{2/3}]^{1/4}} \left[1 + \left(\frac{Re}{282000} \right)^{5/8} \right]^{4/5} \quad [\text{dimensionless}] \quad (14)$$

4 Finding Other Temperatures (At any time t)

Once $T_A(t)$ is found, we can find other temperatures using the quasi-steady-state assumption:

4.1 Surface Temperature $T_B(t)$

$$T_B(t) = T_C + \left(\frac{T_A(t) - T_C}{R_{\text{total}}} \right) R_{\text{env}} \quad [^{\circ}] \quad (15)$$

4.2 Temperature Profile across Insulation $T(r, t)$

This gives the temperature at any radius r (where $r_1 \leq r \leq r_2$) at a specific time t .

$$T(r, t) = T_A(t) - \left[(T_A(t) - T_B(t)) \cdot \frac{(r/r_1)}{(r_2/r_1)} \right] \quad [^\circ] \quad (16)$$

4.3 Soil Temperature Profile $T_{\text{soil}}(r, t)$

This equation allows you to find the temperature at any distance from your cable ($r > r_2$), as long as it is buried.

$$T_{\text{soil}}(r, t) = T_C + (T_B(t) - T_C) \frac{(4z/r)}{(4z/D)} \quad [^\circ] \quad (17)$$

Where z is the burial depth and $D = 2r_2$.

5 Detailed Examples

We will now apply the model to two scenarios. Note that I_{max} (Ampacity) is calculated based on the T_{max} of the insulation. Note on calculations: The values for ρ_e (resistivity) and k_{soil} (conductivity) used in these examples are $\rho_e = 1.68 \times 10^{-8} \Omega$ and $k_{soil} = 1.0^{-1-1}$, respectively, to match the step-by-step verification.

5.1 Example 1: “Household Main” Heavy Load

5.1.1 Given Parameters (Load 1)

- Load: $P = 10\,000$, $V = 230$, $PF = 0.95$
- Conductor (Copper, $\approx 25^2$): $r_1 = 0.00282$, $L = 200$
 $\rho_c = 8960^{-3}$, $c_{p,c} = 385^{-1-1}$, $\rho_e = 1.68 \times 10^{-8} \Omega$
- Insulation (XLPE): $r_2 = 0.004$ ($D = 0.008$), $T_{max} = 90^\circ$
 $\rho_{ins} = 920^{-3}$, $c_{p,ins} = 2150^{-1-1}$, $k_{ins} = 0.28^{-1-1}$
- Environment: $T_C = 20^\circ$

5.1.2 Common Calculations (Load 1)

- $I = 10000/(230 \cdot 0.95) = 45.77$
- $A_c = \pi \cdot (0.00282)^2 = 2.498 \times 10^{-5}^2$
- $R_{elec} = (1.68 \times 10^{-8} \Omega \cdot 200)/(2.498 \times 10^{-5}^2) = 0.1345 \Omega$
- $P_{gen} = (45.77)^2 \cdot 0.1345 \Omega = 281.9$
- $C_{th,c} = (8960^{-3} \cdot \pi \cdot 0.00282^2 \cdot 200) \cdot 385^{-1-1} = 17\,233^{-1}$
- $C_{th,ins} = (920^{-3} \cdot \pi \cdot (0.004^2 - 0.00282^2) \cdot 200) \cdot 2150^{-1-1} = 9998^{-1}$
- $C_{th,corrected} = 17233 + 0.5(9998) = 22\,232^{-1}$
- $R_{ins} = (0.004/0.00282)/(2\pi \cdot 0.28^{-1-1} \cdot 200) = 0.00099^{-1}$

5.1.3 Results by Environment (Load 1)

- Case 1: Underground ($z = 0.7$, $k_{soil} = 1.0^{-1-1}$)
 - $R_{env} = (4 \cdot 0.7/0.008)/(2\pi \cdot 1.0 \cdot 200) = 0.00466^{-1}$
 - $R_{total} = 0.00099 + 0.00466 = 0.00565^{-1}$
 - $T_{ss,heat} = 20 + (281.9 \cdot 0.00565) = 21.59^\circ$
 - $\tau = 0.00565 \cdot 22232 = 125.6$
 - $I_{max} = \sqrt{(90 - 20)/(0.1345 \cdot 0.00565)} = 303.4$
- Case 2: Natural Convection ($v_{wind} = 0^{-1}$)
 - $R_{env} = 0.02438^{-1}$ (from iterative solution)
 - $R_{total} = 0.00099 + 0.02438 = 0.02537^{-1}$
 - $T_{ss,heat} = 20 + (281.9 \cdot 0.02537) = 27.15^\circ$
 - $\tau = 0.02537 \cdot 22232 = 564.0$
 - $I_{max} = \sqrt{(90 - 20)/(0.1345 \cdot 0.02537)} = 143.4$

- Case 3: Forced Convection ($v_{\text{wind}} = 3 \text{ }^{-1}$)
 - $R_{\text{env}} = 0.00312 \text{ }^{-1}$ (from iterative solution)
 - $R_{\text{total}} = 0.00099 + 0.00312 = 0.00411 \text{ }^{-1}$
 - $T_{\text{ss,heat}} = 20 + (281.9 \cdot 0.00411) = 21.16^{\circ}$
 - $\tau = 0.00411 \cdot 22232 = 91.37$
 - $I_{\text{max}} = \sqrt{(90 - 20)/(0.1345 \cdot 0.00411)} = 356.1$

5.2 Example 2: “Washing Machine” Load

5.2.1 Given Parameters (Load 2)

- Load: $P = 2000$, $V = 230$, $PF = 0.65$ (Bad)
- Conductor (Copper, $\approx 2.5^2$): $r_1 = 0.000892$, $L = 200$
(Material properties same as Load 1)
- Insulation (PVC): $r_2 = 0.0017$ ($D = 0.0034$), $T_{\max} = 70^\circ$
 $\rho_{\text{ins}} = 1400^{-3}$, $c_{p,\text{ins}} = 900^{-1-1}$, $k_{\text{ins}} = 0.17^{-1-1}$
- Environment: $T_C = 20^\circ$

5.2.2 Common Calculations (Load 2)

- $I = 2000/(230 \cdot 0.65) = 13.38$
- $A_c = \pi \cdot (0.000892)^2 = 2.499 \times 10^{-6}^2$
- $R_{\text{elec}} = (1.68 \times 10^{-8} \Omega \cdot 200)/2.499 \times 10^{-6}^2 = 1.345 \Omega$
- $P_{\text{gen}} = (13.38)^2 \cdot 1.345 \Omega = 240.8$
- $C_{th,c} = (8960^{-3} \cdot \pi \cdot 0.000892^2 \cdot 200) \cdot 385^{-1-1} = 1724^{-1}$
- $C_{th,ins} = (1400^{-3} \cdot \pi \cdot (0.0017^2 - 0.000892^2) \cdot 200) \cdot 900^{-1-1} = 1660^{-1}$
- $C_{th,\text{corrected}} = 1724 + 0.5(1660) = 2554^{-1}$
- $R_{\text{ins}} = (0.0017/0.000892)/(2\pi \cdot 0.17^{-1-1} \cdot 200) = 0.00302^{-1}$

5.2.3 Results by Environment (Load 2)

- Case 1: Underground ($z = 0.7$, $k_{\text{soil}} = 1.0^{-1-1}$)
 - $R_{\text{env}} = (4 \cdot 0.7/0.0034)/(2\pi \cdot 1.0 \cdot 200) = 0.00534^{-1}$
 - $R_{\text{total}} = 0.00302 + 0.00534 = 0.00836^{-1}$
 - $T_{ss,\text{heat}} = 20 + (240.8 \cdot 0.00836) = 22.01^\circ$
 - $\tau = 0.00836 \cdot 2554 = 21.35$
 - $I_{\max} = \sqrt{(70 - 20)/(1.345 \cdot 0.00836)} = 66.7$
- Case 2: Natural Convection ($v_{\text{wind}} = 0^{-1}$)
 - $R_{\text{env}} = 0.0352^{-1}$ (from iterative solution)
 - $R_{\text{total}} = 0.00302 + 0.0352 = 0.03822^{-1}$
 - $T_{ss,\text{heat}} = 20 + (240.8 \cdot 0.03822) = 29.20^\circ$
 - $\tau = 0.03822 \cdot 2554 = 97.6$
 - $I_{\max} = \sqrt{(70 - 20)/(1.345 \cdot 0.03822)} = 31.2$
- Case 3: Forced Convection ($v_{\text{wind}} = 3^{-1}$)
 - $R_{\text{env}} = 0.00484^{-1}$ (from iterative solution)
 - $R_{\text{total}} = 0.00302 + 0.00484 = 0.00786^{-1}$
 - $T_{ss,\text{heat}} = 20 + (240.8 \cdot 0.00786) = 21.89^\circ$
 - $\tau = 0.00786 \cdot 2554 = 20.07$
 - $I_{\max} = \sqrt{(70 - 20)/(1.345 \cdot 0.00786)} = 68.8$

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B Assumed Material Properties

B.1 Summary of Material Properties

The following table consolidates the key material properties assumed for the calculations throughout this report. These values are representative and based on standard engineering references.

Table 2: Consolidated Material Properties and Assumed Values

Material	Property	Value
Copper	Density (ρ)	8960^{-3}
	Specific Heat (c_p)	385^{-1-1}
	Conductivity (σ)	$56 \Omega^{-1-2}$
	Temp. Coefficient (α)	0.00393^{-1}
Aluminum	Density (ρ)	2700^{-3}
	Specific Heat (c_p)	900^{-1-1}
	Conductivity (σ)	$36 \Omega^{-1-2}$
	Temp. Coefficient (α)	0.00403^{-1}
ACSR	Density (approx. avg.)	3980^{-3}
	Specific Heat (approx. avg.)	880^{-1-1}
	Conductivity (σ)	$34 \Omega^{-1-2}$
PVC	Density (ρ)	1400^{-3}
	Specific Heat (c_p)	1000^{-1-1}
	Thermal Conductivity (k)	0.17^{-1-1}
XLPE	Density (ρ)	920^{-3}
	Specific Heat (c_p)	2300^{-1-1}
	Thermal Conductivity (k)	0.28^{-1-1}
Soil	Thermal Resistivity (ρ_{soil})	1.5^{-1}

C Beyond the Lumped Model: A PDE Approach

The Lumped Capacitance Model (ODE) is an excellent engineering approximation, but it relies on a key simplification: it treats the insulation as a "massless" resistor and lumps all thermal capacitance into the conductor. This Appendix briefly explains the "real" physics model, which uses a Partial Differential Equation (PDE), and its connection to Fourier analysis.

C.1 The "Real" Physics: The Heat Diffusion Equation

A more realistic model treats the insulation as a region that *both* resists and *stores* heat. This means the temperature T is a function of *both* radius r and time t , written as $T(r, t)$. This is governed by the Heat Diffusion Equation, which in cylindrical coordinates (assuming only radial flow) is:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \left[\frac{1}{3} \right] \quad (18)$$

Where:

- Left Side: Represents the net heat conduction *into* a small ring of insulation.
- Right Side: Represents the rate of heat *stored* in that small ring of insulation.
- $\alpha = \frac{k_{\text{ins}}}{\rho_{\text{ins}} c_{p,\text{ins}}}$: is the thermal diffusivity of the insulation, a material property that measures how quickly heat diffuses relative to how much it stores.

This PDE cannot be solved by simple integration. It must be solved simultaneously with a set of complex boundary conditions (e.g., balancing conductor heat generation/storage with conduction into the insulation) and an initial condition ($T(r, 0) = T_i$).

C.2 The Solution Strategy: Fourier Series and Separation

The classic method to solve this PDE is Separation of Variables.

1. Assume a Solution Form: We assume the solution can be "separated" into a product of two functions, one for space and one for time:

$$T(r, t) = R(r) \cdot \Theta(t)$$

2. Split the PDE: Plugging this assumption into the PDE and rearranging, we can split the one complex PDE into two simpler ODEs:

- An ODE for the spatial part, $R(r)$.
- An ODE for the time part, $\Theta(t)$.

3. Solve the Time ODE: The time ODE, $\Theta(t)$, solves to a simple exponential decay, just like our lumped model: $\Theta(t) = C(-\lambda t)$.

4. Solve the Spatial ODE (The Fourier Connection): The spatial ODE, $R(r)$, is more complex. Its solutions are not simple functions, but a special set of functions called Bessel functions.

Just like a complex musical chord can be represented as a Fourier Series (a sum of simple sines and cosines), our complex initial temperature profile can be represented as a Fourier-Bessel Series (a sum of these Bessel functions).

Each of these functions represents a "thermal mode" or a "shape" of heat distribution.

C.2.1 Putting It All Together

The "real" solution $T(r, t)$ is an infinite sum (a Fourier-Bessel series) of these spatial shapes, where each shape decays in time at its own unique rate:

$$T(r, t) = \sum_{n=1}^{\infty} A_n R_n(r) (-\lambda_n t)$$

Our simple lumped model (with its single τ) is essentially a first-order approximation of this. It uses only the *first and most dominant term* ($n = 1$) of this infinite series, effectively capturing the slowest, most dominant time constant of the system.

C.2.2 Fourier Transforms vs. Series

- Fourier Series (and their cousins, Fourier-Bessel series) are used when the problem has finite boundaries, like our insulation layer (from r_1 to r_2).
- Fourier Transforms are the tool used when the problem has infinite boundaries. For example, a more advanced model for the *underground* case could use a Fourier Transform to analyze the diffusion of heat from the cable into an infinitely large volume of soil.